

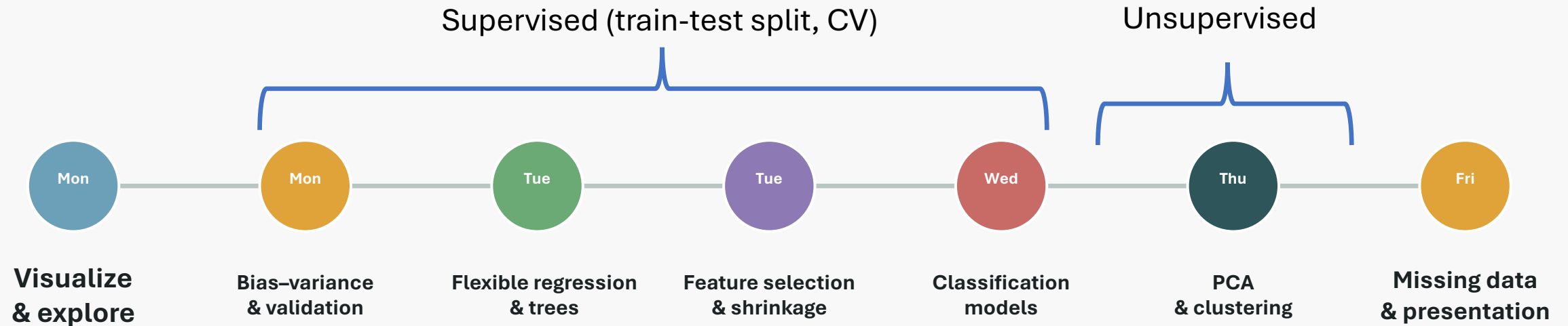
From Data to Decisions

Bringing Everything Together

Mahdi Shafiee Kamalabad

Department of Methodology, Statistics and Data Science
Utrecht University

From data to decisions: the course journey



Core idea

Machine learning is not only “running an algorithm”; it is a sequence of choices about data, features, models, validation, and interpretation.

Practical focus

The course used R and real examples to make methods understandable and useful in research practice.

Final task

Use your analysis to tell a clear story: question → method → evidence → conclusion.

How to choose a method: start with the question

What is the RQ and the goal?

Known outcome Y?

No outcome label?

Predict a number

Regression: linear models, polynomial/spline regression, regression trees. Evaluate with test MSE or cross-validation.

Predict a class

Classification: logistic regression, LDA/QDA, trees, random forests, boosting, SVM. Evaluate with accuracy, sensitivity/specificity, ROC/AUC.

Reduce dimensions

PCA: summarize many correlated variables into a few components; use scores, loadings, and explained variance.

Find groups

Clustering: k-means or hierarchical clustering; choose distance, scaling, and number of clusters carefully.

Did I separate training and test data? Did I control overfitting?

From question to useful insight

A strong data-science analysis is more than choosing an algorithm.

1	Define the question	What is the goal: explain, predict, classify, reduce dimensions, or discover groups?
2	Understand the data	Visualize variables and relationships; inspect missingness patterns, outliers, imbalance, and data quality.
3	Prepare the data	Clean and transform variables; handle missing data carefully; encode categories; scale when needed; prevent information leakage.
4	Select the approach	Match the method to the goal and outcome: regression, classification, PCA, or clustering.
5	Fit the method & tune settings	Fit the chosen method and choose settings carefully: for supervised learning tune model complexity; for unsupervised learning choose settings such as K, distance, linkage, scaling, or number of components.
6	Evaluate honestly	Use unseen data and predictive metrics for supervised learning; for unsupervised learning, assess structure, stability, interpretability, and sensitivity to analytical choices.
7	Check robustness & Interpret the result	Explain and interpret coefficients, loadings, clusters, importance, stability, and limitations.
8	Communicate & act	Connect question → method → evidence → conclusion, and state what the result supports in practice.

Course materials: https://mshafieek.github.io/IntroTo_ML_DataAnalysis/